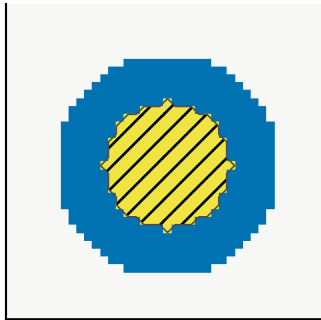
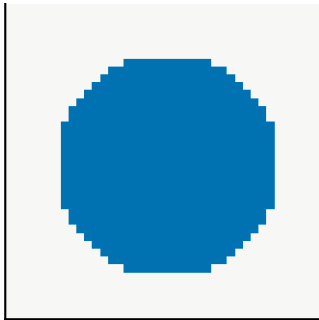


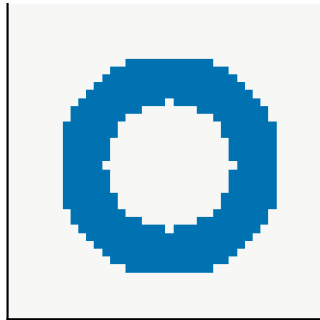
(a) ring over POSSIBLE
 $\beta_1(\text{CERT_FG}) = 1$ — but fillable



(b) in-budget adversary fills the interior — cycle destroyed



(c) ring over CERT_BG
unfillable — certified



 CERT_FG (certainly grout)

 POSSIBLE (adversary's)

 CERT_BG (certainly bg)